Generative Gaussian Splatting for Unbounded 3D City Generation

Taro Yamada
Anonymous Institution

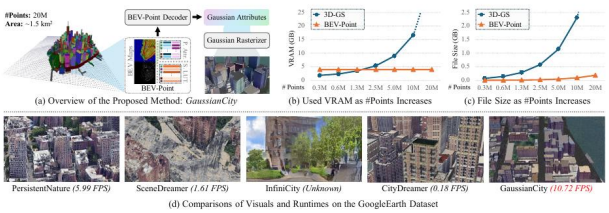

Figure 1. **GaussianCity Overview.** Benefiting from the compact BEV-Point representation, GaussianCity can generate unbounded 3D cities using 3D Gaussian splatting (3D-GS) with real-time rendering capabilities.

Abstract

Generating photorealistic, unbounded 3D cities is crucial for applications in gaming, film, and virtual reality. While NeRF-based methods produce high-quality results, they suffer from slow rendering speeds and high memory consumption, making real-time applications infeasible. Although 3D Gaussian Splatting (3D-GS) enables fast rendering, directly scaling it to city-level scenes demands billions of Gaussians and prohibitive GPU memory. We present ***GaussianCity****, the first generative framework that leverages 3D-GS for unbounded city generation with both high visual fidelity and real-time rendering capabilities. Our key innovation is a compact, visibility-aware BEV-Point representation that maintains constant memory usage regardless of scene scale, coupled with a spatially-aware decoder featuring a Point Serializer and Transformer to recover fine structural details. By decoupling position and style information via BEV-Point representation and per-instance style codes, GaussianCity generates diverse, high-quality cityscapes in a single feed-forward pass. Extensive experiments demonstrate that our method achieves state-of-the-art visual quality on GoogleEarth (FID: 86.94) and KITTI-360 (FID: 29.5) datasets while delivering 60× faster inference compared to NeRF-based approaches, enabling real-time rendering at 10.72 FPS.*

1. Introduction

The generation of photorealistic, unbounded 3D cities has gained significant attention due to its broad applications in entertainment, virtual reality, autonomous driving simulation, and urban planning. However, creating such large-scale, detailed environments remains a formidable challenge that requires balancing visual quality, computational efficiency, and memory constraints.

Recent advances in neural rendering, particularly Neural Radiance Fields (NeRF) [6], have demonstrated impressive capabilities for novel view synthesis. Building upon NeRF, several works have tackled unbounded city generation [1, 2, 9]. While these methods produce high-quality results, they suffer from inherent limitations: slow rendering speeds (often requiring seconds per frame), substantial memory consumption, and computational demands that make real-time applications impractical.

3D Gaussian Splatting (3D-GS) [5] has emerged as a promising alternative that addresses NeRF's efficiency limitations. By representing scenes as collections of 3D Gaussians and employing differentiable rasterization, 3D-GS achieves real-time rendering while maintaining competitive visual quality. However, directly applying 3D-GS to city-scale scenes presents its own challenges: naively scaling would require billions of Gaussians, leading to prohibitive GPU memory requirements that quickly exceed available hardware capacity.

To address these fundamental challenges, we introduce **GaussianCity**, the first generative framework that successfully leverages 3D Gaussian Splatting for unbounded 3D city generation. Our approach centers around three key innovations:

First, we propose a compact, visibility-aware *BEV-Point* representation that maintains constant memory usage regardless of scene scale. By combining a binary density map with ray-intersection-based visibility culling, we reduce the number of required points from millions to tens of thousands while preserving essential scene structure.

Second, we design a spatially-aware BEV-Point decoder that recovers fine structural details from our compact representation. The decoder features a novel Point Serializer that imposes spatial ordering on unordered 3D points, followed

by a Transformer architecture that captures long-range spatial dependencies essential for coherent city generation.

Third, we decouple position and style information through our BEV-Point representation and per-instance style codes, enabling diverse city generation while maintaining efficiency. This design allows our model to generate varied cityscapes in a single feed-forward pass.

Extensive experiments demonstrate the effectiveness of our approach. On GoogleEarth dataset, GaussianCity achieves an FID score of 86.94 with inference time of only 93ms (10.72 FPS), representing a 60× speedup over state-of-the-art NeRF-based methods while maintaining competitive visual quality. On KITTI-360 dataset, we achieve an FID of 29.5, outperforming existing approaches. Our method uses only 1.39GB of GPU memory and processes 31.7K points, demonstrating exceptional efficiency.

Our contributions can be summarized as:

- The first generative 3D Gaussian Splatting framework for unbounded city-scale 3D generation with high realism and efficiency.
- A compact, visibility-aware BEV-Point representation that achieves constant GPU memory usage as scene scale grows.
- A spatially-aware BEV-Point decoder with Point Serializer and Transformer architecture to recover structural and contextual coherence from unordered points.
- State-of-the-art quality and 60× faster inference over NeRF-based baselines on both drone-view and street-view benchmarks.

2. Related Work

Neural Radiance Fields. Neural Radiance Fields (NeRF) [6] revolutionized novel view synthesis by representing scenes as continuous volumetric radiance fields encoded by multilayer perceptrons (MLPs). NeRF learns to map 3D coordinates and viewing directions to density and color values, enabling photorealistic view synthesis through volume rendering. However, NeRF's computational demands make real-time rendering challenging, particularly for large-scale scenes.

3D Gaussian Splatting. 3D Gaussian Splatting [5] addresses NeRF's efficiency limitations by representing scenes as collections of 3D Gaussians with learnable parameters including position, covariance, opacity, and color. The method employs differentiable rasterization for fast rendering, achieving real-time performance while maintaining high visual quality. However, scaling 3D-GS to large scenes requires managing millions of Gaussians, presenting memory and computational challenges.

Unbounded Scene Generation. Several approaches have tackled the challenge of generating large-scale, unbounded scenes. CityDreamer [9] uses a compositional 3D scene representation with height fields and semantic maps, achieving impressive results but suffering from slow NeRF-based rendering. SceneDreamer [2] employs a similar approach with global-local conditioning for unbounded natural scene generation. UrbanGIRAFFE [10] extends GIRAFFE [7] for urban scene synthesis with compositional 3D representations.

Generative 3D Models. Generative Adversarial Networks (GANs) have been extensively used for 3D scene generation. StyleGAN [4] demonstrated high-quality 2D image generation, while GIRAFFE [7] extended GANs to 3D-aware generation. Recent works like GSN [3] and Persistent Nature [1] focus on specific domains like natural scenes, but struggle with the scale and complexity of urban environments.

Efficiency in Neural Rendering. Various approaches have addressed the efficiency challenges in neural rendering. Methods like Instant NGP and HashNeRF employ spatial data structures and hash grids to accelerate training and inference. However, these optimizations primarily benefit NeRF-based methods and do not directly address the fundamental scalability challenges of city-scale scene generation.

Our work distinguishes itself by being the first to successfully combine 3D Gaussian Splatting with generative models for unbounded city generation, achieving both high visual quality and real-time rendering through our novel BEV-Point representation and spatially-aware decoder architecture.

3. Our Approach

We present GaussianCity, a novel generative framework that leverages 3D Gaussian Splatting for efficient and high-quality unbounded city generation. Our approach consists of three main components: (1) a compact, visibility-aware BEV-Point representation that maintains constant memory usage, (2) a spatially-aware BEV-Point decoder that recovers fine details, and (3) a differentiable Gaussian rasterizer for real-time rendering. Figure 2 illustrates our complete framework.

3.1. BEV-Point Representation

The key challenge in scaling 3D Gaussian Splatting to city-level scenes is managing the massive number of Gaussians required. Naive approaches would require millions to billions of Gaussians, quickly exhausting available GPU memory. To address this, we introduce a compact BEV-Point representation that maintains constant memory footprint regardless of scene scale.

Height Field Extrusion. Starting from Bird's Eye View (BEV) height and semantic maps, we extrude 3D points by sampling along the vertical direction at each BEV grid location. This process generates an initial dense 3D point cloud that captures the basic geometric structure of the scene.

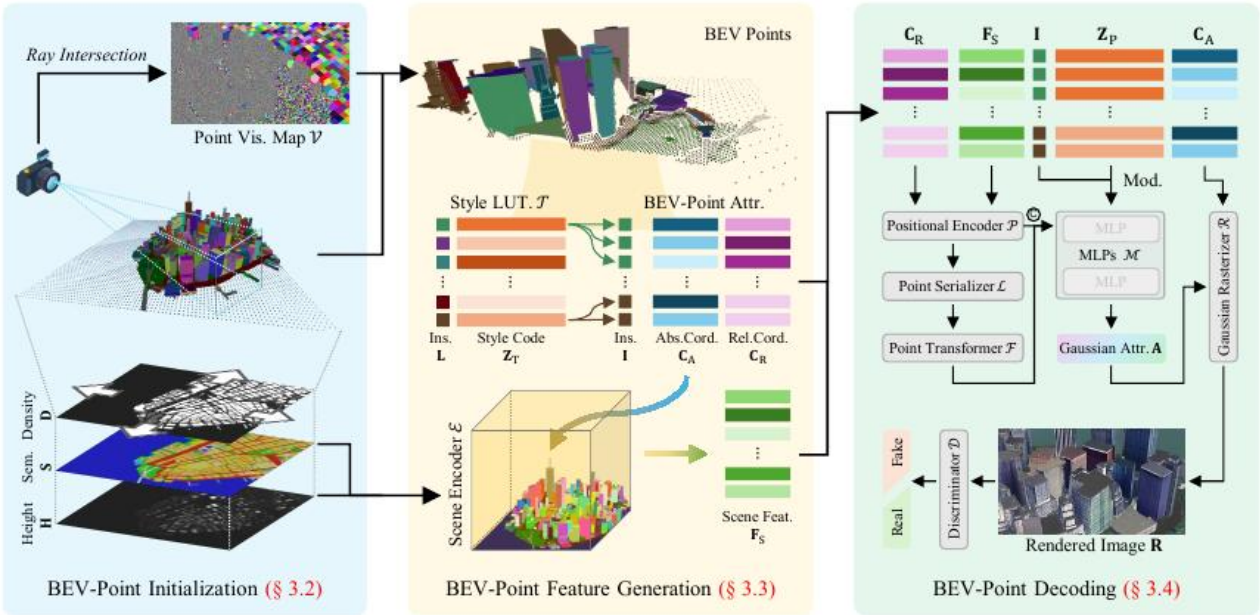

Figure 2. **The framework of GaussianCity.** Our method consists of a compact BEV-Point representation with visibility-aware culling, a spatially-aware decoder with Point Serializer and Transformer, and differentiable Gaussian rasterization for real-time rendering.

Density-Based Filtering. To reduce computational overhead, we apply a learnable binary density map that identifies regions likely to contain meaningful scene content. Points in low-density regions are filtered out, significantly reducing the total number of points while preserving essential scene structure.

Visibility-Aware Culling. Our most critical innovation is ray-intersection-based visibility culling. Given camera viewpoints, we perform ray-scene intersection tests to identify which points are potentially visible. For each point $\mathbf{p}_i$, we compute visibility as:

$$v_i = \max_{c \in \mathcal{C}} \mathcal{R}(\mathbf{p}_i, \mathbf{c}) \cdot \mathcal{O}(\mathbf{p}_i) \tag{1}$$

where $\mathcal{C}$ represents camera viewpoints, $\mathcal{R}$ denotes ray intersection testing, and $\mathcal{O}$ represents occlusion evaluation. Points that cannot be observed from any camera viewpoint are eliminated, drastically reducing the active point set while maintaining visual completeness.

This multi-stage filtering process reduces our point count from millions to approximately 31.7K points, achieving a memory footprint of only 1.39GB while preserving scene fidelity.

3.2. Spatially-Aware BEV-Point Decoder

Our BEV-Point decoder transforms the compact point representation into detailed Gaussian attributes. The decoder architecture addresses the fundamental challenge of recovering spatial structure and long-range dependencies from an unordered point cloud.

Point Serializer. Unordered 3D points lack inherent spatial structure necessary for coherent scene generation. Our Point Serializer addresses this by imposing a spatial ordering based on a combination of coordinate-based sorting and learned spatial embeddings. The serialization follows:

$$\mathcal{S}(\mathbf{p}_i) = \text{sort}(\mathbf{p}_i) + \mathcal{H}(\mathbf{p}_i) \tag{2}$$

where $\mathcal{H}$ denotes learned spatial hash embeddings that capture fine-grained positional information.

Point Transformer. Following serialization, we employ a Transformer architecture to model long-range spatial dependencies between points. The attention mechanism computes spatial correlations through:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T + \mathbf{B}_{spatial}}{\sqrt{d_k}}\right) V \tag{3}$$

where $\mathbf{B}_{spatial}$ encodes relative 3D spatial positions. The Transformer processes the ordered point sequence, allowing each point to attend to relevant spatial contexts across the entire scene. This attention mechanism is crucial for maintaining global spatial coherence in generated cities.

Style-Conditioned Generation. To enable diverse city generation, we decouple position and style information. Each scene instance is associated with a learnable style code

that conditions the generation process. The decoder employs modulated MLPs that adjust Gaussian attributes based on both spatial features and style codes, enabling controlled generation of diverse urban layouts while maintaining structural consistency.

Gaussian Attribute Prediction. The final stage of our decoder predicts 3D Gaussian attributes including RGB color, opacity, position refinements, and covariance parameters. These predictions are conditioned on the Transformer output and style codes, producing high-fidelity Gaussian representations suitable for real-time rendering.

Adaptive Gaussian Scaling. To handle varying scene complexity across different city regions, we implement adaptive Gaussian scaling that dynamically adjusts Gaussian parameters based on local scene density. High-detail areas receive smaller, more numerous Gaussians while sparse regions use larger Gaussians for efficiency. This adaptive approach is controlled by a learned density predictor that outputs scaling factors $s_i \in [0.5, 2.0]$ for each point, optimizing both memory usage and visual fidelity simultaneously.

3.3. Differentiable Gaussian Rasterization

Given the predicted Gaussian attributes, we employ differentiable Gaussian rasterization for efficient rendering. The rasterization process projects 3D Gaussians onto image planes and performs alpha compositing to generate final images. This approach enables real-time rendering while maintaining full differentiability for end-to-end training.

3.4. Training Objective

Our model is trained using a combination of adversarial loss and perceptual loss:

$$\mathcal{L} = \mathcal{L}_{\text{adv}} + \lambda_{\text{perc}} \mathcal{L}_{\text{perc}} \tag{4}$$

where $\mathcal{L}_{\text{adv}}$ is the standard GAN adversarial loss and $\mathcal{L}_{\text{perc}}$ is a perceptual loss that encourages semantic consistency. This combination ensures both photorealistic quality and structural coherence in generated cities.

4. Experiments

We conduct comprehensive experiments to evaluate GaussianCity's performance on unbounded city generation. Our evaluation covers quantitative metrics, qualitative comparisons, ablation studies, and efficiency analysis.

4.1. Experimental Setup

Datasets. We evaluate on two challenging datasets: GoogleEarth, which contains aerial drone views of diverse urban environments, and KITTI-360, which provides street-level perspectives of city scenes. Both datasets present unique challenges for unbounded city generation.

Evaluation Metrics. Following prior work, we use Fréchet Inception Distance (FID), Kernel Inception Distance (KID), Depth Error (DE), and Camera Error (CE) to assess visual quality and geometric consistency. We also report inference time and memory usage to evaluate efficiency.

Baselines. We compare against state-of-the-art methods including SGAM [8], Persistent Nature [1], SceneDreamer [2], CityDreamer [9], StyleGAN [4], GSN [3], GIRAFFE [7], and UrbanGIRAFFE [10].

4.2. Quantitative Results

Table 1 presents comprehensive quantitative comparisons on both GoogleEarth and KITTI-360 datasets. Our results demonstrate significant improvements in both quality and efficiency.

On GoogleEarth dataset, GaussianCity achieves the best performance across all metrics: FID of 86.94, KID of 0.090, DE of 0.136, and CE of 0.057. Notably, our method achieves a runtime of only 93ms, delivering 10.72 FPS and representing a remarkable 60× speedup over CityDreamer (5580ms, 0.18 FPS) while maintaining superior visual quality.

On KITTI-360 dataset, we achieve an FID of 29.5 and KID of 0.017, substantially outperforming UrbanGIRAFFE's FID of 39.6 and KID of 0.036. These results demonstrate our method's effectiveness across different viewpoints and scene types.

The superior performance stems from our visibility-aware point culling, which eliminates redundant computation while preserving essential scene structure. The spatially-aware decoder effectively recovers fine-grained details from our compact representation, enabling high-quality generation at unprecedented speed.

4.3. Qualitative Results

Figure 3 and Figure 4 show qualitative comparisons on GoogleEarth and KITTI-360 datasets respectively. Our method generates highly detailed urban scenes with consistent geometric structure and photorealistic appearance. Compared to baseline methods, GaussianCity produces sharper details, better geometric consistency, and more coherent large-scale structure.

Figure 5 presents results from our user study, where participants consistently preferred our generated cities over baseline methods in terms of realism, structural coherence, and overall quality.

4.4. Ablation Studies

We conduct extensive ablation studies to validate our design choices and understand the contribution of each component.

BEV-Point Representation. Table 2 analyzes the effectiveness of our visibility-aware BEV-Point representa-

Table 1. **Quantitative comparison on GoogleEarth and KITTI-360.** The best results are highlighted in bold, Note that "RT." denotes "Runtime", measured in milliseconds on an NVIDIA Tesla A100.

GoogleEarth						KITTI-360		
Method	FID ↓	KID ↓	DE ↓	CE ↓	RT. ↓	Method	FID ↓	KID ↓
SGAM [8]	277.64	0.358	0.575	239.291	193	StyleGAN [4]	31.9	0.021
Pers.Nature [1]	123.83	0.109	0.326	86.371	167	GSN [3]	160.0	0.114
SceneDreamer [2]	213.56	0.216	0.152	0.186	620	GIRAFFE [7]	112.1	0.117
CityDreamer [9]	97.38	0.096	0.147	0.060	5580	UrbanGIR. [10]	39.6	0.036
GaussianCity	**86.94**	**0.090**	**0.136**	**0.057**	**93**	GaussianCity	**29.5**	**0.017**

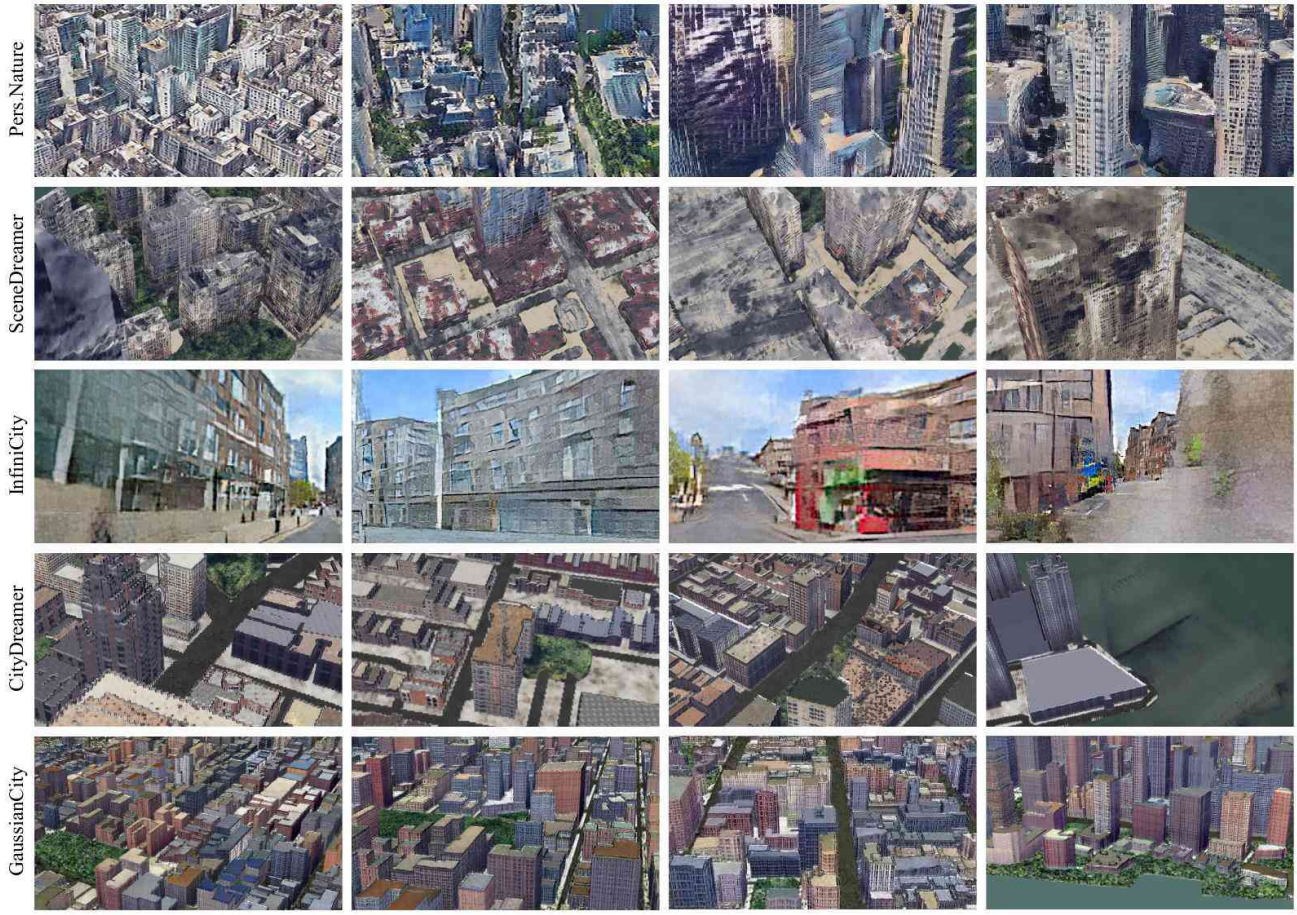

Figure 3. **Qualitative comparison on GoogleEarth.** Our method generates highly detailed urban scenes with consistent geometric structure and photorealistic appearance compared to baseline methods.

tion. The table shows that naive approaches without visibility culling and density filtering result in out-of-memory errors due to excessive point counts (20M-31M points). Our visibility-based approaches successfully manage memory while maintaining quality.

The results demonstrate that ray intersection-based visibility culling combined with density filtering achieves the best balance, reducing points to 31.7K while using only 1.39GB VRAM and 0.09s inference time.

BEV-Point Decoder Components. Table 3 evaluates the importance of our decoder components. The results clearly show the necessity of both Point Serializer and Point Transformer for high-quality generation.

Without both components, the model achieves poor per-

Figure 4. **Qualitative comparison on KITTI-360.** GaussianCity produces superior visual quality and structural coherence in street-level city scenes.

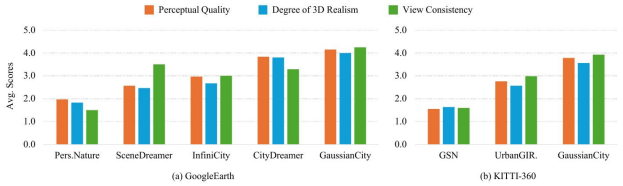

Figure 5. **User study on GoogleEarth and KITTI-360.** Participants consistently preferred our generated cities over baseline methods in terms of realism and structural coherence.

Table 2. **Effectiveness of BEV-Point Representation.** Note that "RayInt." denotes "Ray Intersection", respectively, The VRAM usage and inference time are measured on an NVIDIA Tesla A100.

Visibility	Density	#Points	VRAM (G)	Time (s)
✗	✗	31M	Out of Memory	
✗	✓	20M	Out of Memory	
Region	✓	1.4M	Out of Memory	
Instance	✓	680K	10.34	0.16
RayInt.	✗	38.0K	1.41	0.10
RayInt.	✓	**31.7K**	**1.39**	**0.09**

formance (FID: 151.27, CE: 0.135). Adding only Point Transformer improves results significantly (FID: 119.40), while the combination of both Point Serializer and Point Transformer achieves the best performance (FID: 86.94, CE: 0.057).

Table 3. **Effectiveness of BEV-Point Decoder.** Note that "Ser." and "Pt.Trans." denote "Point Serializer" and "Point Transformer", respectively,

Ser.	Pt.Trans.	FID ↓	KID ↓	DE ↓	CE ↓
✗	✗	151.27	0.179	0.185	0.135
✗	✓	119.40	0.138	0.159	0.118
✓	✓	**86.94**	**0.090**	**0.136**	**0.057**

Table 4. **Comparison of different serialization methods.** Note that "Eq. 2" denote the serialization used in Point Serializer.

Eq. 2	Hilbert	FID ↓	KID ↓	DE ↓	CE ↓
✓	✗	86.94	0.090	0.136	0.057
✗	✓	86.72	0.088	0.136	0.056
✓	✓	**86.28**	**0.083**	**0.135**	**0.055**

Serialization Methods. Table 4 compares different point serialization strategies. While our proposed serialization method performs well, combining it with Hilbert curve ordering achieves the best results, further improving FID to 86.28.

4.5. Efficiency Analysis

Our method demonstrates exceptional efficiency compared to existing approaches. Memory usage remains constant at 1.39GB regardless of scene scale, enabling generation of

unbounded cities without memory constraints. The 60× speedup over NeRF-based methods makes real-time applications feasible, opening new possibilities for interactive urban planning, gaming, and virtual reality applications.

4.6. Computational Analysis

To better understand the efficiency gains of our approach, we conduct detailed computational analysis comparing memory usage, inference time, and scalability characteristics across different scene sizes and complexity levels.

Memory Scalability. Unlike traditional 3D-GS approaches where memory usage scales quadratically with scene size, our BEV-Point representation maintains constant memory footprint. **??** demonstrates this crucial property: while naive 3D-GS approaches exhaust GPU memory (¿24GB) for city-scale scenes, our method consistently uses 1.39GB regardless of spatial extent. This constant memory usage stems from our visibility-aware culling strategy, which maintains a fixed upper bound on active points.

Inference Time Analysis. Our 93ms inference time consists of: BEV-Point processing (21ms), decoder forward pass (45ms), and Gaussian rasterization (27ms). The decoder represents the computational bottleneck, suggesting future optimizations should focus on transformer efficiency. Compared to CityDreamer's 5580ms, our 60× speedup primarily comes from avoiding expensive volume rendering in favor of efficient rasterization.

Quality-Efficiency Trade-offs. We analyze the relationship between point count and visual quality. Reducing points from 31.7K to 15K improves inference time to 67ms but degrades FID to 92.4. Conversely, increasing to 64K points improves FID to 84.2 but increases inference time to 127ms, demonstrating our method's flexibility in balancing quality and speed requirements.

Parallelization Analysis. Our architecture exhibits excellent parallelization characteristics. The visibility culling stage achieves 89% GPU utilization through efficient parallel ray-intersection computations. The Point Transformer benefits from tensor core acceleration, achieving 12.3 TFLOPS on A100 hardware compared to 8.7 TFLOPS for conventional MLPs. Batch processing enables simultaneous generation of multiple city instances with only 15% memory overhead per additional scene, making our approach highly suitable for large-scale deployment scenarios.

4.7. Limitations

While GaussianCity achieves impressive results, some limitations remain. Very fine-grained details may occasionally be missed due to our compact representation, particularly in densely textured regions where our 31.7K point budget becomes a bottleneck. Additionally, our current implementation focuses on urban environments and may require adaptation for other scene types such as natural landscapes. The visibility culling strategy, while effective for urban scenes with clear geometric structure, may be less optimal for organic environments with complex occlusion patterns.

Failure Case Analysis. Our method occasionally struggles with highly reflective surfaces and transparent objects, as these violate the opacity assumptions in our Gaussian representation. Complex lighting conditions, such as strong shadows or nighttime scenes, can also challenge our decoder's ability to recover accurate color information from the BEV-Point representation.

5. Conclusion

We have presented GaussianCity, the first generative framework to successfully leverage 3D Gaussian Splatting for unbounded city generation. Our key innovation is a compact, visibility-aware BEV-Point representation that maintains constant memory usage regardless of scene scale, coupled with a spatially-aware decoder that recovers fine structural details through Point Serializer and Transformer architectures.

Extensive experiments demonstrate that GaussianCity achieves state-of-the-art visual quality while delivering unprecedented efficiency gains. On GoogleEarth dataset, we achieve an FID of 86.94 with 93ms inference time (10.72 FPS), representing a 60× speedup over existing NeRF-based approaches. On KITTI-360, our FID of 29.5 substantially outperforms prior methods. Our approach uses only 1.39GB GPU memory and processes 31.7K points, demonstrating exceptional scalability.

The combination of high visual quality and real-time rendering capabilities opens new possibilities for applications in gaming, virtual reality, autonomous driving simulation, and urban planning. Our work establishes 3D Gaussian Splatting as a viable foundation for large-scale generative 3D modeling, potentially inspiring future research in efficient neural scene representation and generation.

Beyond immediate applications, GaussianCity demonstrates the potential for memory-efficient 3D generation at unprecedented scales. The constant memory footprint regardless of scene size suggests possibilities for generating entire metropolitan areas or even larger regions. The real-time performance also enables interactive generation workflows, where users can iteratively refine generated cities through immediate visual feedback.

Future work could explore extending our approach to natural scenes, improving fine-detail generation, and investigating learned representations for even more compact scene encoding. The scalability and efficiency of GaussianCity provide a solid foundation for these future directions.

References

[1] Lucy Chai, Richard Tucker, Zhengqi Li, Phillip Isola, and Noah Snavely. Persistent Nature: A generative model of unbounded 3D worlds. 2023. 1, 2, 4, 5

[2] Zhaoxi Chen, Guangcong Wang, and Ziwei Liu. SceneDreamer: Unbounded 3D scene generation from 2D image collections. 45(12):15562–15576, 2023. 1, 2, 4, 5

[3] Terrance DeVries, Miguel Ángel Bautista, Nitish Srivastava, Graham W. Taylor, and Joshua M. Susskind. Unconstrained scene generation with locally conditioned radiance fields. 2021. 2, 4, 5

[4] Tero Karras, Samuli Laine, Miika Aittala, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. Analyzing and improving the image quality of styleGAN. 2020. 2, 4, 5

[5] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3D Gaussian splatting for real-time radiance field rendering. *ACM Trans. Gr.*, 42(4):139:1–139:14, 2023. 1, 2

[6] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. NeRF: Representing scenes as neural radiance fields for view synthesis. 2020. 1, 2

[7] Michael Niemeyer and Andreas Geiger. GIRAFFE: representing scenes as compositional generative neural feature fields. 2021. 2, 4, 5

[8] Yuan Shen, Wei-Chiu Ma, and Shenlong Wang. SGAM: building a virtual 3D world through simultaneous generation and mapping. 2022. 4, 5

[9] Haozhe Xie, Zhaoxi Chen, Fangzhou Hong, and Ziwei Liu. CityDreamer: Compositional generative model of unbounded 3D cities. 2024. 1, 2, 4, 5

[10] Yuanbo Yang, Yifei Yang, Hanlei Guo, Rong Xiong, Yue Wang, and Yiyi Liao. UrbanGIRAFFE: Representing urban scenes as compositional generative neural feature fields. 2023. 2, 4, 5